

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BL3)		
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Section / Batch:	slot-DN 2024	Date:	11/08/26.

Code of Conduct

I K. Manoj Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10

1.Explain the concept of image enhancement in the spatial domain. Discuss how basic gray-level transformations, histogram processing techniques, and spatial filtering can be used to improve the quality of the captured images. Justify the suitability of each technique for this scenario.

2.Differentiate between image smoothing and image sharpening. Explain the working principles of spatial smoothing filters and sharpening filters. Discuss how these techniques improve image quality and explain the situations in which each technique should be applied.

3.Explain the image enhancement process in the frequency domain. Describe the roles of low-pass filters and high-pass filters in image enhancement. Compare their effects on image quality and discuss suitable applications for each filtering technique.

4.Explain the concept of image segmentation. Compare point detection, line detection, edge detection, thresholding, and region-based segmentation with respect to their working principles, advantages, limitations, and practical applications. Identify the most appropriate technique(s) for this scenario and justify your answer.

5.Explain the importance of edge detection, edge linking, and boundary detection in image processing. Discuss the working principles of these techniques and describe how they contribute to accurate object recognition and defect detection. Illustrate your answer with suitable examples and diagrams wherever necessary.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
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Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Introduction:

Image Enhancement is the process of improving the visual quality of an image so that important details become more visible.

1. Basic Gray-level Transformation.

It modify the intensity values of pixels.

$$\frac{23}{25}$$

fr

Types:

a) Image negative.

converts dark regions into bright regions and vice versa.

$$s = L - 1 - r$$

r = input pixel value

L = number of gray levels.

b) log transformations.

expands dark pixel values

compresses bright pixel values

formula

$$s = c \log(1+r)$$

c) Power law: controls image brightness

$\gamma < 1$: brighter image

$\gamma > 1$: dark image

$$S = c r^\gamma$$

Histogram processing:

Shows distribution of gray levels.

Histogram Equalization:

* Redistributes fixed intensities.

* Improves contrast automatically

Spatial filtering:

uses neighboring pixels to process an image

* Smoothing filters remove noise

* Sharpening filters enhance edges and details.

Conclusion:

Spatial domain enhancement improves image quality
Histogram processing enhances contrast.

Image Smoothing is Image Sharpening.

Image Smoothing

Removes noise

Produces a blurred image

Uses low-pass filters

Image Sharpening

Enhances edges.

Produces a sharper image

Uses high-pass filters

Smoothing filters:

Replace pixels with the average or median of neighboring pixels.

ex: mean filter, median filter, Gaussian filter.

Sharpening filters:

Enhance intensity differences b/w neighboring pixels.

ex: Laplacian, Sobel, Prewitt

Applications:

Smoothing: medical images, noisy photographs.

Sharpening: OCR, fingerprint, industrial inspection.

Frequency Domain

is an image processing technique in which an image is first transformed from the spatial domain

low pass filter.

high pass filter

Band pass filter

<u>LPP</u>	<u>HPP</u>
Removes noise smooths image	Enhances edge Sharpens image

★ low-pass filter

slow intensity change

Ex: 10 11 12 13 14

Brightness changes slowly
pixels all almost similar.

★ high pass filter

fast intensity change

Ex: 10 10 10 200 200

Sudden jump in brightness

★ Band-pass filter.

very fast & random change

10 2 30 15 240 8.

Image Segmentation:

divides an image into meaningful regions or objects, making it easier to analyze.

- * Separate objects from background.
- * Identify Regions of Interest.
- * Enable object recognition and measurement.

Types

Threshold-Based Segmentation

Edge-Based Segmentation

Region-Based Segmentation

Clustering-Based Segmentation

Deep learning-Based Segmentation.

Point Detection: Detects isolated pixels.

Line Detection: Detects straight lines.

Edge Detection: Detects boundaries.

Technique	Advantages	Limitations
Point Detection	Simple	Sensitive to noise
Line Detection	Clear Accurate	missed curved objects
Thresh. Edge Detection	clear Boundaries	noise sensitive
Thresholding	fast	Poor under uneven lighting
Region-Based Segmentation	Accurate	computationally expensive.

- * medical image analysis
- * face recognition
- * OCR.

Conclusion:

Segmentation helps identify objects accurately.
 Thresholding is suitable for simple images,
 while region based is best for complex image.

Edge Detection.

Finds Sudden changes in image intensity

Operators, Sobel, prewitt, canny

* used to detect object edges.

Edge linking.

Connects broken changes edge pixels to form Continuous edges

* Improves object shape representation.

Boundary Detection:

Extracts the outer contour of an Image. object

* used for measuring object size and shape

Applications:

Object recognition

crack and defect detection.

medical imaging.

Conclusion.

Edge detection identifies boundaries, edge linking connects them, boundary detection extracts the outline of object, improves object recognition